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Research Topic: Risk Indicators and Pharmacist-Led Monitoring Strategies for Predicting Pneumonia-to-Sepsis Progression in Elderly Insulin-Dependent Diabetic Patients: A Systematic Literature Review.

Title: Risk Indicators and Pharmacist-Led Monitoring Strategies for Predicting Pneumonia-to-Sepsis Progression in Elderly Insulin-Dependent Diabetic Patients: A Systematic Literature Review.

Research Problem

Community-acquired pneumonia is one of the major global health problems at present, with mortality rates rising both with age and in patients with comorbidities such as diabetes mellitus [2]. According to WHO, the number of people living with diabetes worldwide rose from 200 million in 1990 to 830 million in 2022, with prevalence rising fastest in low- and middle-income countries [1]. Of those living with diabetes, approximately 96 percent have type 2 diabetes [2]. Diabetes patients face a 1.4 times higher risk of getting pneumonia than the general population and up to 5.9 times higher risk of invasive pneumococcal disease [2].

Hyperglycemia heightens the level of risk in the immune system. Chronic high blood glucose can impair neutrophil chemotaxis, reduce complement activation, interfere with phagocytosis, and delay T-cell-mediated immune responses, which can weaken the body's ability to fight infection [2]. In patients with insulin-dependent diabetes, this immune dysfunction can be more pronounced, making it more difficult for the body to clear respiratory infections. When pneumonia develops in diabetic patients, uncontrolled blood glucose levels can contribute to fluid and electrolyte imbalances, mitochondrial dysfunction, and oxidative stress and in severe cases can lead to multiple organ failure and sepsis [3].

Vaccination provides meaningful protection against infection, but it does not offer complete protection. One randomized controlled trial reported pneumococcal vaccine efficacy of up to 89% among immunocompetent older adults with diabetes, although the biological reason for this variation remains unclear [2]. This highlights an important gap in current care: even vaccinated patients with insulin dependent diabetes may develop severe breakthrough infection. Once an infection occurs, however, few reliable tools currently exist to identify which patients are at risk of progressing to sepsis before their condition becomes critical.

This research was motivated by the case of my grandmother, a 68-year-old woman with a five-year history of insulin-dependent diabetes mellitus who received her pneumococcal vaccine annually. Approximately six weeks prior to her passing, she developed pneumonia, with early symptoms including odynophagia (painful swallowing) and dizziness. Despite being vaccinated, her condition progressed to community-acquired pneumonia. Within two days of acute decline, she developed severe diarrhea and significant dehydration, followed by rapid clinical deterioration

accompanied by rigors. She subsequently required life support, with a declining pulse that did not respond to initial medication, requiring a change to an alternative agent. She also developed oliguria, indicating progressive kidney dysfunction. Her condition progressed to septic shock with multiple organ failure, including neurological failure resulting in brain death. Despite resuscitation efforts, she passed away. Her case demonstrates how rapidly a vaccinated patient with insulin-dependent diabetes can deteriorate when secondary complications occur, particularly when early symptoms may present atypically and go unrecognized. It also highlights the broader gap addressed by this research: the lack of a consolidated framework for identifying and flagging patients at high risk of rapid deterioration before their condition becomes critical.

Existing Literature

The association between diabetes mellitus and the severity of pneumonia has been well documented in the literature. Compared with the general population, individuals with diabetes have a higher risk of developing pneumococcal pneumonia and invasive pneumococcal disease, which is largely associated with impaired immune function resulting from chronic hyperglycemia [2]. Research at the cellular level has further demonstrated that prolonged hyperglycemia can impair neutrophil activity, skew macrophage polarization, and alter T-cell-mediated immune responses. Together, these changes can reduce the body's ability to effectively recognize and eliminate bacterial infections [3].

Beyond pneumonia, diabetes is also associated with an increased susceptibility to bacterial infections. High blood glucose levels can create conditions that support bacterial growth, while diabetes-related immune dysfunction can impair neutrophil chemotaxis and the production of bactericidal free radicals that are important for eliminating pathogens [4]. These findings suggest that the poorer outcomes observed in diabetic patients with pneumonia are part of a wider vulnerability to infection rather than being limited to pneumonia alone.

Although the mechanisms linking diabetes with infection severity are relatively well understood, there is limited research on how these findings can be applied to clinical prediction. One randomized controlled trial reported substantial differences in pneumococcal vaccine efficacy between elderly patients with and without diabetes, although the biological reasons for this difference remain unclear [2]. This highlights a gap between evidence on population-level vaccine protection and the ability to

identify individual diabetic patients who may remain particularly vulnerable to severe infection.

Interventions aimed at improving glycemic control have also shown promising results in hospitalized patients. For example, one study found that clinical pharmacist-led glycemic control teams, using structured insulin protocols and patient education, improved glycemic management among patients admitted to infectious disease wards [5]. This finding aligns with broader evidence that pharmacist-led approaches improve glycemic management in hospitalized patients more generally [5]. However, these interventions have not been specifically examined among elderly insulin-dependent diabetic patients with pneumonia who may be at risk of progression to sepsis.

Overall, the existing literature provides strong evidence linking diabetes, hyperglycemia, immune dysfunction, and increased infection severity. It also suggests that pharmacist-led glycemic monitoring can contribute to better outcomes in hospitalized patients. However, these areas of research have largely been examined separately. There remains a need for a more integrated approach that considers factors such as glycemic control, hydration status, and vaccination history when assessing the risk of sepsis in insulin-dependent diabetic patients with pneumonia. The potential role of pharmacist-led monitoring within such an approach also remains insufficiently explored.

Research Question

Which clinical and pharmacological risk indicators - including glycemic control, hydration status, and vaccination history - best predict progression from community-acquired pneumonia to sepsis in elderly insulin-dependent diabetic patients, and what pharmacist-led monitoring strategies could enable earlier intervention?

Methodology

This research will be conducted as a systematic literature review using peer-reviewed studies to identify clinical and pharmacological factors associated with the progression of pneumonia to sepsis in elderly patients with insulin-dependent diabetes. Relevant studies will be searched in PubMed/MEDLINE and the Cochrane Library. Google Scholar will also be used as a supplementary source to find studies that may not be indexed in the main databases.

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The main search will use the following terms: ("insulin-dependent diabetes" OR "type 1 diabetes" OR "insulin therapy") AND ("pneumonia" OR "community-acquired

pneumonia") AND ("sepsis" OR "septic shock") AND ("elderly" OR "older adults"). Additional searches will focus on specific areas related to the review, including "glycemic control AND infection outcomes," "pneumococcal vaccine effectiveness diabetic," and "pharmacist-led glycemic monitoring."

Studies will be included if they involve patients aged 65 years or older with insulin-dependent diabetes and examine pneumonia or its progression to sepsis or septic shock. The review will be limited to studies published within the past 10–15 years. Peer-reviewed research articles, systematic reviews, and cohort studies will be considered. Studies involving children or non-diabetic populations, as well as studies focused on type 2 diabetes without insulin therapy, will be excluded. Case reports, anecdotal reports, studies not related to pneumonia-associated sepsis, and studies that are not available in English or cannot be reliably translated will also be excluded.

For each eligible study, information such as sample size, identified risk factors, odds ratios and confidence intervals, mortality or disease progression rates, and outcomes of relevant interventions will be recorded. The extracted information will be compared across the included studies and presented in a table. Common risk factors found across multiple studies will then be examined to identify patterns. These findings will be used to develop a risk-flagging framework for identifying elderly insulin-dependent diabetic patients with pneumonia who may be at higher risk of progressing to sepsis.

Research Topic Justification

13 Sepsis is a major cause of death among elderly diabetic patients who are hospitalized with pneumonia. Compared with non-diabetic patients with the same infection, diabetic patients are more likely to require ICU care, have longer hospital stays, and experience higher 30-day and in-hospital mortality [2]. Since sepsis can worsen quickly once it develops, identifying patients at higher risk as early as possible is important for timely treatment. This is particularly relevant for insulin-dependent diabetic patients, who can still develop severe infections even after vaccination, as shown by the case that led to this research.

Although diabetes-related pneumonia has been widely studied, the existing research is spread across different areas. Factors such as vaccination, blood glucose control, hydration, and pharmacist-led monitoring have generally been studied separately. There is limited research bringing these factors together specifically for elderly
8 insulin-dependent diabetic patients with pneumonia. Therefore, the main focus of this research is not to identify a completely new risk factor, but to bring together existing evidence and use it to develop a single risk-flagging framework for this patient group.

The research may also have practical value in healthcare settings. Pharmacists are already part of many hospital care teams and can play a role in monitoring blood glucose and managing medication-related issues. Previous research has also shown benefits from pharmacist involvement in the management of patients with infections and diabetes. A risk-flagging framework based on the findings of this review could therefore provide a basis for further research into pharmacist-led monitoring of high-risk patients. Overall, the study aims to connect existing evidence on diabetes and infection with a practical approach for identifying patients who may need closer monitoring.

Quality of Writing

This proposal uses precise, academically appropriate language directed at a multidisciplinary audience with minimal jargon. Grammar, citation formatting, and structure follow IEEE style and align with IARCO's submission guidelines throughout.

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